

conditional statements

interview questions

Q1: What is the purpose of conditional statements in Python?

- Conditional statements in Python are used to execute different blocks of code based on whether a given condition is true or false. This allows the program to make decisions and follow different execution paths depending on the conditions evaluated.

Q2: How is indentation used in Python, especially in the context of conditional statements?

- In Python, indentation is crucial as it defines the scope of code blocks. Indentation is used to indicate which statements belong to a specific conditional block. For instance, the code inside an `if` statement must be indented to signify that it should be executed if the condition is true.

Q3: How do you write an `if-else` statement, and when is it used?

```
if condition:
    # code block to execute if condition is true
else:
    # code block to execute if condition is false
```

The `else` statement is used to provide an alternative block of code that executes when the condition in the `if` statement is false.

Q4: Explain the `if-elif-else` structure in Python with an example.

```
if condition1:
    statement1
elif condition2:
    statement2
elif condition3:
```

```
    statement3
else:
    statement4
```

Q5: What is a nested `if` statement, and when would you use it?

- A nested `if` statement is an `if` statement inside another `if` or `else` block. It is used for more complex decision-making processes where multiple levels of conditions need to be evaluated.

```
if condition1:
    if condition2:
        # code block for condition2 being true
    else:
        # code block for condition2 being false
else:
    # code block for condition1 being false
```

Q6: What happens if the condition in an `if` statement is false and there is no `else` statement?

- **A8:** If the condition in an `if` statement is false and there is no `else` statement, the block of code inside the `if` statement is skipped, and the program continues to execute the subsequent lines of code.

Q7: Explain the difference between `if` statements and `elif` statements.

- **A10:** An `if` statement is used to evaluate a single condition. If the condition is true, the corresponding block of code is executed. An `elif` (else if) statement is used to check additional conditions if the preceding `if` or `elif` condition(s) are false. It allows for multiple conditions to be evaluated sequentially, providing multiple branches of execution.